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United States Patent Application Publication

20250282211

Kind Code

A1

Publication Date

September 11, 2025

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SELF POWERING SHAFT

Abstract

A device and method for improving the efficiency of internal combustion engine driven vehicles and other machines using a supplemental electric drive power source on a drive shaft with an electric motor and generator placed in series. As speed of the vehicle increases, the speed of the driveshaft will increase, producing more power at the driveshaft via the electric motor and generator assembly. As the generator produces more power, the electric motor will have more power going to it until a desired speed is achieved. The desired speed will be maintained by the wheels turning the rear axle therein turning the generator and the electric motor. This will allow the vehicle to maintain highway speeds at the same rpm but with less load causing an exponential increase in fuel efficiency as the device and method shall supplement required vehicle force/power output. A device and method for improving the efficiency of a primary power supply with a combination motor/generator with a common drive shaft.

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Family ID: 96948562

Appl. No.: 19/217720

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation-in-part 18447521 20230810 ABANDONED child US 19217720

Publication Classification

Int. Cl.: B60K6/485 (20071001); B60K6/26 (20071001)

U.S. Cl.:

Background/Summary

[0001] This application is a Continuation-in-part of application Ser. No. 18/447,521 (the '521 application) filed Aug. 10, 2023, co-owned with the present invention. The '521 application is incorporated here by reference for all purposes, and with respect to all of which the present invention claims priority under United States Patent Law.

BACKGROUND OF THE INVENTION

[0002] The invention relates to devices and methods for improving fuel efficiency in internal combustion driven motor vehicles.

[0003] The standard automobile power system generally consists of a motor powered by internal combustion, where the force of fossil fuel is compressed and ignited, by which it pushes down a piston, thus turning a crankshaft, therefore turning a flywheel. From the flywheel power is transferred to a transmission, where the gearing ratio determines the speed and force by which the power is routed out of the transmission to a drive shaft, then to the rear end gearing, turning the wheels.

[0004] In today's modern world, the search for the answer to the problems of expendable fossil fuels and uses thereof have eluded us. Our dependence on such fuels is an environmental and economic strain. The immediate resolve would be to stop using fossil fuels, unfortunately this is not a feasible option due to the lack of technology to replace fossil fuel motors and still get the desired performance. The object of the proposed invention is to improve fuel efficiency and longevity for the standard automobile.

SUMMARY OF THE INVENTION

[0005] The object of the present invention herein shall be referred to as a Self-Powering Shaft ("SPS") to improve fuel efficiency in most existing rear wheel and some front wheel and four-wheel drive vehicles. The inventive Self-Powering Shaft does not replace any major drive components (motor, transmission) of the standard automobile, with installation on current vehicles being relatively easy to integrate along the existing drive shaft of a rear wheel drive train assembly. The Self-Powering Shaft can also be used in many other applications. The Self-Powering Shaft modifies the drive shaft of an existing rear wheel drive train assembly by adding at least one electric generator and at least one electric motor, concentrically rotating by mechanical connections to the drive shaft both with and on the same axis of rotation as the drive shaft. The vehicle's drive shaft that runs from the rear of the transmission to the rear differential will be driven to rotate and drive the electric generator and will then be turned and driven by the electric motor. The forward momentum of the vehicle and wheels rotating with the rear axle shall keep the drive shaft spinning, therein turning the electric generator powering the electric motor. This will take a majority of the load off the motor, reducing required internal combustion motor rpm. The load of an internal combustion motor is the main culprit in fuel deficiency. For example, when driving up a steep hill, the MPG of a vehicle drops dramatically. However, when going downhill the MPG jumps exponentially.

[0006] The type of electric motor and generator is not limited to one specific type. The components of the present invention can also be placed inside of a drive axle housing. For the purposes of explanation, the example used will be the drive shaft of a rear-wheel-drive vehicle.

[0007] Each electric generator will be wired and electrically connected to power at least one electric motor affixed to the drive shaft so that when at speed, the generator shall power at least one motor to assist in rotating the drive shaft and reducing the load on the vehicle's internal combustion motor, thereby reducing the fuel consumed to produce such rotation. A voltage regulator and switch

will be installed between the wiring of the generator and electric motor to adjust the motor's voltage as the vehicle slows, causing the electric motor to produce less power and to totally stop voltage when the electric motor stops. If the regulator switches polarity, the motor will reverse in the direction of the drive shaft assisting in slowing the vehicle. It is a further object of the inventive Self-Powering Shaft to interface with integral engine control modules.

[0008] Another huge advantage of the Self-Powering Shaft is its ability to be added to tractor trailers or semi-trucks to allow the trailer to pull its own weight or at a minimum assist in the forward motion of the vehicle and slowing. The Self-Powering Shaft can be added to most progressive vehicles used today inclusive of older models, marine, farm and industrial applications.

[0009] In an another embodiment of the present invention the Self-Powering Shaft can be configured to comprise a combination motor/generator intertwined in one casing with one set of brushes and commutator for the motor and another set for the generator, with the combination motor/generator being rotationally affixed to a rotating drive shaft directly driven by a primary power source such as an internal combustion engine or an electric motor used to power a vehicle or other machine. In this embodiment the Self-Powering Shaft would recapture energy generated by the primary power source lost to momentum or inefficiencies in the power train and return it to power the rotating drive shaft thereby increasing the efficiency of the power source. For example, the forward momentum of an internal combustion engine powered automobile would power the Self-Powering Shaft and reduce the load on the internal combustion engine. Likewise, the rotational momentum of a power driven drive shaft configured as the Self-Powering Shaft would reduce the load on the power source, whether it be an internal combustion engine, or an electric motor, or any source of rotational power. The Self-Powering Shaft can only function with a primary power source. The generator function cannot power the motor function without rotation of the drive shaft initiated by the primary power source.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 depicts a plan view of the SPS concept affixed to the drive shaft of a vehicle's rear wheel drive train assembly, with the addition of a plurality of electric generators and electric motors concentrically mounted on the drive shaft.

[0011] FIG. 2-2 is a cross sectional view of the front electric motor concentrically mounted on the drive shaft.

[0012] FIG. 3-3 is a cross sectional view of the front electric generator concentrically mounted on the drive shaft.

[0013] FIG. 4 is an exploded view of a combination motor/generator shown without a casing with a common driven shaft.

DETAILED DESCRIPTION OF THE INVENTION

[0014] In FIG. 1 a plan view of the drive shaft 2 of an existing rear wheel drive train assembly is depicted with the addition of at least one combination of electric generators 7 and 9 and electric motors 6 and 8 concentrically mounted on the drive shaft 2 to rotate by mechanical connection to the drive shaft 2, with front U-joint 3 and rear U-joint 4, running from the transmission 1 to the rear differential 5. Also shown is an electrical connection 10 between electric generator 7 and electric motor 6, and an electrical connection 11 between electric generator 9 and electric motor 8. A control box 12 shall be electrically connected by electrical connection 19 to the generators and motors at electrical connections 10 and 11 where the control box 12 contains at least one voltage regulator 20 and switch 21 for each combination of electric generators and motors. The voltage regulator 20 will adjust the voltage as the vehicle slows causing the electric motor to produce less power, with a switch 21 to totally stop voltage to the motor when slowing the vehicle. If the

regulator **20** switches polarity from the generator to the motor, it will cause the motor to try to spin in the opposite direction of the drive shaft **2** causing slowing of the vehicle by assisting in the braking of the vehicle. Also shown is an engine control module **22** interfaced with the control box **12**.

[0015] FIGS. **2-2** and FIG. **3-3** show cross sectional views of the front electric motor **6** and the front electric generator **7** respectively, with each concentrically mounted to the drive shaft **2** by mechanical connection so that the motor armature **13** and the generator armature **14** will rotate with the shaft **2**. The drive shaft **2** will become the driving shaft for the front electric motor **6** and the front electric generator **7**. Also shown in FIG. **2-2** is the motor armature **13**, the motor outer frame **15** and a motor mount **17** for securing the electric motor **6** to a cross member of the vehicle's frame whilst permitting movement of the drive shaft **2** as it interfaces and functions with the rear differential **5** and suspension. FIG. **3-3** shows the generator armature **14**, the generator outer frame **16** and a generator mount **18** for bolting the electric generator **7** to a frame cross member of the vehicle to allow movement of the drive shaft **2** as it relates to the rear differential **5** and suspension. The minimal movement of the driveshaft **2** will have a minimal effect on the components of the motor **6** and generator **7** that touch, such as brushes.

[0016] FIG. **4** is an exploded view of a combination motor/generator **21** intertwined in one casing not shown with one set of brushes **24** and commutator **23** for the motor and another set **29** and **30** for the generator, with the combination motor/generator **21** being rotationally affixed to a rotating common driven shaft **22** directly driven by a primary power source such as an internal combustion engine or an electric motor used to power a vehicle or other machine. Referring to FIG. **1**, there would also be an electrical connection **10** between the motor and generator functions within the casing, as well as a control box **12** with switch **20** and voltage regulator **21**. The control box **12** with switch **20** and voltage regulator **21** are essential to prevent detrimental results in certain applications. For example, in the loss of primary power, the momentum of the driven shaft would continue to power the combination motor/generator **21** unless switched off.

[0017] Not every application of the present invention will be installed in an identical fashion. In some cases, the existing driveshaft can be used to rotate a motor or motors and a generator or generators by concentric mounting to the driveshaft, which is to say, the axis of rotation of the driveshaft is the same for the motor or motors and a generator or generators. Alternatively, the motor or motors and a generator or generators could rotate on an axis parallel to the axis of rotation of the driveshaft with power transferred through a belt or a chain. In others, the drive shaft will need to be replaced or the present invention installed on an alternative power drive receiving power from an internal combustion driven motor. In some situations, the device will need to be installed similar to a transfer case, in others it will fasten to the vehicle's rear end or differential **5**. The idea of the invention is to use an electric generator and motor in conjunction with combustion of fossil fuels to reduce the load of the internal combustion engine, thereby increasing fuel economy.

[0018] While initially envisioned for improving the efficiency of internal combustion driven automobiles, the Self-Powering Shaft with combination motor/generator **21** being rotationally affixed to a rotating drive shaft **22** can be directly driven by any primary power source used to power a vehicle or other machine.

Claims

1. A device for installation on an existing vehicle driven by an internal combustion motor alone with a drive shaft to improve fuel efficiency of the internal combustion motor, comprising: at least one electric generator rotating by mechanical connection to the drive shaft; at least one electric motor rotating by mechanical connection to the drive shaft; an electric connection between the electric generator and the electric motor; a control box connected electrically to the electric connection between the electric generator and the electric motor, and the control box further

comprising at least one voltage regulator and switch.

2. The device of claim 1 further comprising an internal combustion motor control module to interface with the control box.

3. The device of claim 2 where the electric generator is concentrically mounted on the drive shaft to rotate by mechanical connection to the drive shaft and the electric motor is concentrically mounted on the drive shaft to rotate by mechanical connection to the drive shaft.

4. A device for installation on an existing vehicle driven by an internal combustion motor alone with an alternative power drive receiving power from the internal combustion motor to improve the fuel efficiency of the internal combustion motor comprising: at least one electric generator rotating by mechanical connection to the power drive from the internal combustion motor; at least one electric motor rotating by mechanical connection to the power drive from the internal combustion motor; an electric connection between the electric generator and the electric motor; a control box connected electrically to the electric connection between the electric generator and the electric motor, and the control box further comprising at least one voltage regulator and switch.

5. The device of claim 4 further comprising an integral combustion motor control module to interface with the control box.

6. A method for installation of a device on an existing vehicle driven by an internal combustion motor alone with a drive shaft to improve the fuel efficiency of the internal combustion motor, comprising: provide at least one electric generator to rotate by mechanical connection to the drive shaft; provide at least one electric motor to rotate by mechanical connection to the drive shaft; provide an electric connection between the electric generator and the electric motor so that the electric generator will power the electric motor to assist in rotating the drive shaft to reduce the load on the internal combustion motor; electrically connect a control box to the electric connection between the electric generator and the electric motor, and where the control box further comprises at least one voltage regulator and switch, and; provide an integral combustion motor control module to interface with the control box.

7. The method of claim 6 where the provided electric generator is concentrically mounted on the drive shaft to rotate by mechanical connection to the drive shaft and the provided electric motor is concentrically mounted on the drive shaft to rotate by mechanical connection to the drive shaft.

8. A method for installation of a device on an existing vehicle driven by an internal combustion motor alone with an alternative power drive receiving power from the internal combustion motor to improve the fuel efficiency of the internal combustion motor comprising: providing at least one electric generator rotating by mechanical connection to the power drive from the internal combustion driven motor vehicle; providing at least one electric motor rotating by mechanical connection to the power drive from the internal combustion driven motor vehicle; providing an electric connection between the electric generator and the electric motor; electrically connect a control box to the electric connection between the electric generator and the electric motor, and where the control box further comprises at least one voltage regulator and switch, and; provide an internal combustion motor control module to interface with the control box.

9. A device for improving the efficiency of a primary power source with a rotational drive shaft, comprising: a combination motor/generator with motor function and generator functions within a casing, rotationally affixed to the drive shaft; an electrical connection between the function and generator functions; a control box connected electrically to the electrical connection between the function and generator functions, and the control box further comprising at least one voltage regulator and switch.

10. The device of claim 9 where the primary power source is an internal combustion engine.

11. The device of claim 9 where the primary power source is an electrical motor.

12. A method for improving the efficiency of a primary power source with a rotational drive shaft, comprising: provide a combination motor/generator with motor function and generator functions within a casing; provide an electrical connection between the function and generator functions;

provide a control box connected electrically to the electrical connection between the function and generator functions, and the control box further comprising at least one voltage regulator and switch; and rotationally affix the combination motor/generator to the drive shaft.
